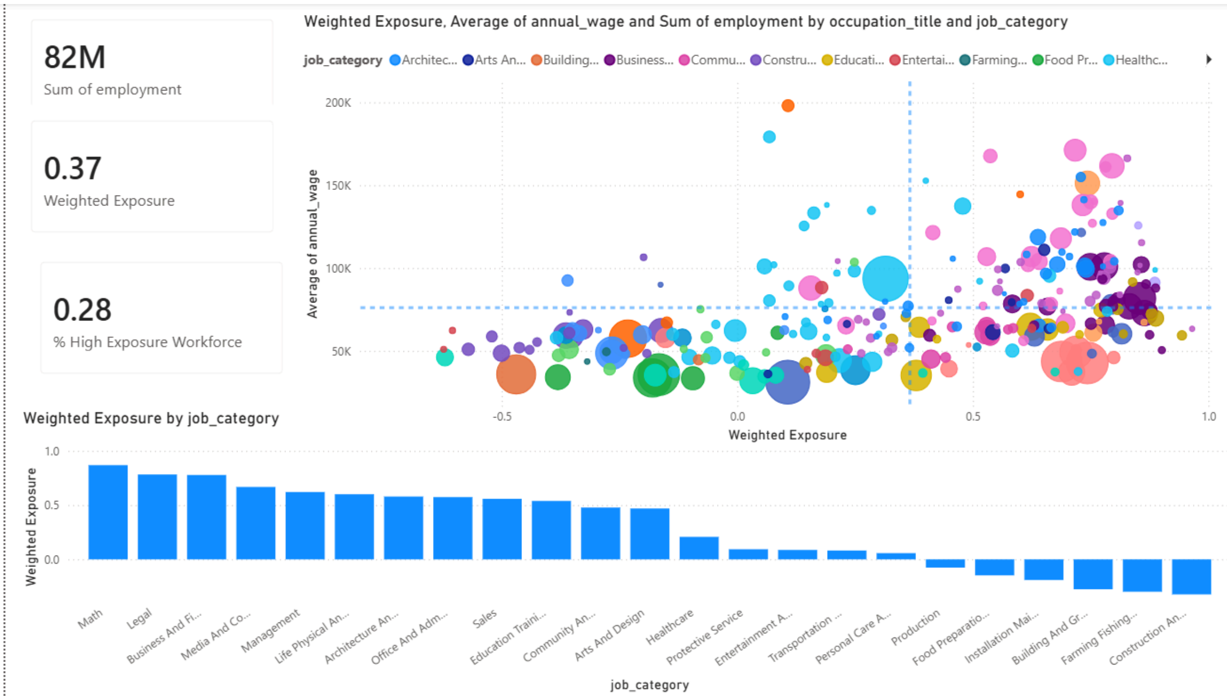
The AI Exposure Landscape: Analyzing Workforce Risk

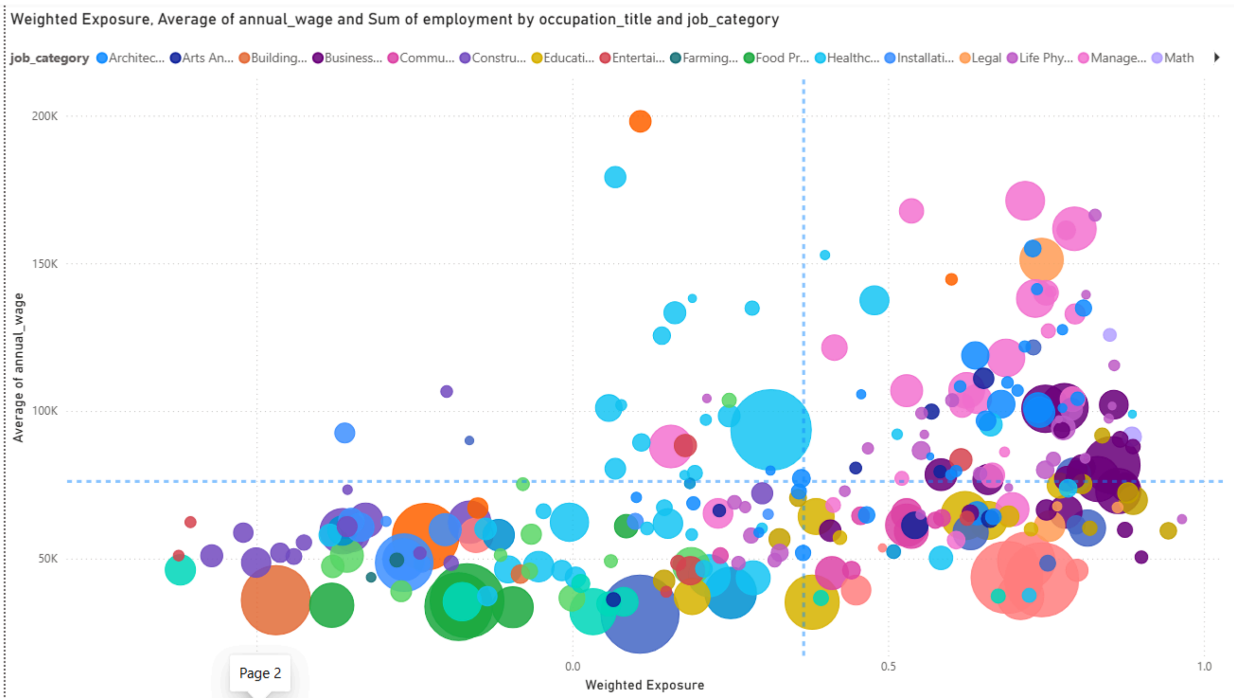


Executive Summary

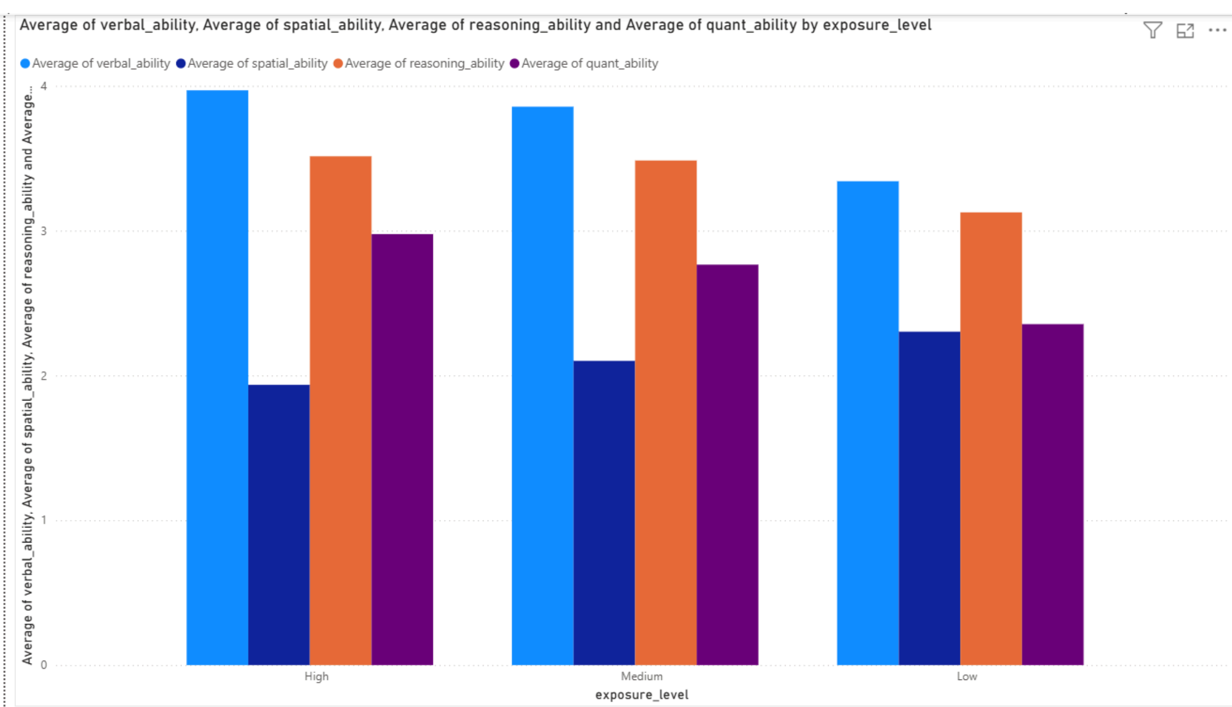
For this project, I wanted to figure out exactly which jobs, industries, and education levels are most at risk from AI. Instead of just looking at one data source, I combined older AI exposure data with newer LLM-based metrics to build a more balanced, realistic view of where the workforce is actually heading.

Key Findings

- **White-Collar Jobs are the Most Exposed:** The data shows that knowledge work is right in the crosshairs. The Math and Media & Communication sectors have the highest average AI exposure at 0.58 and 0.56. On the flip side, physical labor is very safe, Construction & Extraction (0.06) and Farming & Forestry (0.09) are the least exposed industries.
- **The College Degree Paradox:** Past waves of automation usually hit lower-education jobs the hardest, but AI is doing the exact opposite. Jobs that require a Bachelor's degree have the highest AI exposure rate at 0.47, putting almost 24.9 million jobs at risk. Jobs that require no formal education have an exposure rate of just 0.12.

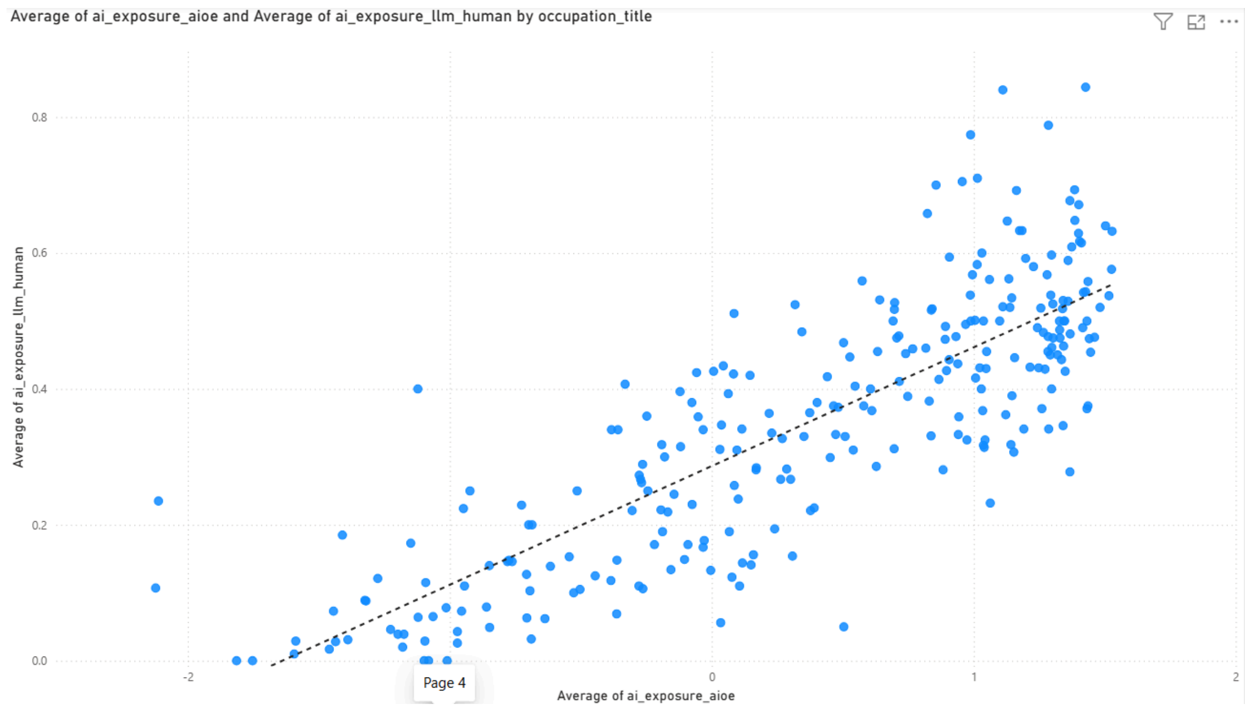


- **High Salary Doesn't Mean High Security:** Earning a lot of money doesn't automatically protect you from AI anymore. When I mapped this out, I found some interesting splits:
  - **High Wage / High Risk:** Computer & Information Systems Managers ($171,200) and Physicists ($166,290).
  - **High Wage / Low Risk:** Airline Pilots ($198,100) and Dentists ($179,210).
- **AI Wants Your Desk Job, Not Your Toolbelt:** AI is clearly targeting specific cognitive skills. The most highly exposed jobs heavily require Verbal (3.97) and Reasoning (3.51) skills. However, Spatial Ability is a massive defense against AI, it is the only cognitive skill that scores higher in low-exposure jobs (2.30) than in high-exposure jobs (1.94).



Methodology & Bias Check

I specifically chose not to trust just one AI model for this analysis. When I plotted the older pre-LLM scores against the newer LLM scores, they mostly agreed on desk jobs, but they disagreed wildly on physical and creative jobs. For example, Fitness Trainers and Dancers had massive divergence scores over 2.2. Averaging these scores together helped smooth out those biases and made the final data model much more reliable.



Final Takeaway

If we want to future-proof careers, we need to stop pushing purely quantitative and verbal skills. The safest jobs of the future are going to rely heavily on spatial reasoning, physical work, and complex human interaction.